RACE Regular Analysis Through Continuous Exercise

Monotonicity : RACE-01

Single Correct Question

1. $f(x) = |x-1| + 2|x-3| - |x+2|$ is monotonically increasing at $x =$?

(A) – 2 (B) 0 (C) 3 (D) 5

MMN125

2. For $0 \le x \le 1$, the function $f(x) = |x| + |x-1|$ is-

(A) Monotonically increasing (B) Monotonically decreasing

(C) Constant function (D) Identity function

MMN126

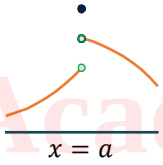
3. The function $y = \frac{2x^2-1}{x^4}$ is

(A) Always increasing (B) Always decreasing

(C) Neither increasing nor decreasing (D) None of these

MMN127

4. In the following graph comment on monotonicity of $f(x)$ at $x = a$.



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(A) Monotonically increasing at $x = a$

(B) Monotonically decreasing at $x = a$

(C) neither monotonically increasing nor decreasing at $x = a$

(D) Can't say at $x = a$

MMN128

5. When $x \in (0,1)$, function $f(x) = 1/\sqrt{x}$ is -

(A) Increasing (B) Decreasing

(C) Neither increasing nor decreasing (D) Constant

MMN129

6. Let f be the function $f(x) = \cos x - \left(1 - \frac{x^2}{2}\right)$ then $f(x)$ is strictly increasing in the interval

(A) $(-\infty, \infty)$ (B) $(-2, \infty)$ (C) $[0, \infty)$ (D) $(0, \infty)$

MMN130

7. Function $f(x) = \frac{e^{2x}-1}{e^{2x}+1}$ is-

(A) Increasing (B) Decreasing

(C) Neither increasing nor decreasing (D) Even function

MMN131

8. Function $f(x) = \ell n \sin x$ is monotonically increasing when-

(A) $x \in (\pi/2, \pi)$ (B) $x \in (-\pi/2, 0)$ (C) $x \in (0, \pi)$ (D) $x \in (0, \pi/2)$

MMN132

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9. For what value of $'a'$ the function $f(x) = x + cos\ x - a$ increases

(A) 0 (B) 1 (C) –1 (D) Any value

MMN133

10. Let $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$ be an increasing function in the set of real numbers R. Then a & b satisfy the condition:

(A) $a^2 - 3b - 15 > 0$ (B) $a^2 - 3b + 15 \leq 0$

(C) $a^2 + 3b - 15 < 0$ (D) $a > 0 \,\&\, b > 0$

MMN134

Multiple Correct Question

11. Let $f(x) = \begin{cases} x & x \in [1,2] \\ 5-x & x \in (2,4) \\ 2 & x = 4 \\ 7-x & x \in (4,6] \end{cases}$ then which of the following statement is/are correct about $f(x)$?

(A) Function is strictly increasing at point $x = 2$

(B) Function is strictly increasing at point $x = 4$

(C) Function is not increasing at point $x = 2$ and $x = 4$

(D) Function is not increasing at point $x = 1$

MMN135

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	D	C	C	C	B	C	A	D	D	B	A,B

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RACE Regular Analysis Through Continuous Exercise

Monotonicity : RACE-02

Single Correct Question

1. The greatest, the least values of the function, $f(x) = 2 - \sqrt{1 + 2x + x^2}, x \in [-2, 1]$ are respectively

(A) 2, 1 (B) 2, –1 (C) 2, 0 (D) –2, 3

MMN136

2. Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $f(x)$. As b varies, the range of $m(b)$ is

(A) $[0,1]$ (B) $\left(0, \frac{1}{2}\right]$ (C) $\left[\frac{1}{2}, 1\right]$ (D) $(0,1]$

MMN137

3. Which of the following is true

(A) $1 + x^2 > (x\sin x + \cos x)$ for $x \in [0, \infty)$ (B) $1 + x^2 \geq (x\sin x + \cos x)$ for $x \in [0, \infty)$

(C) $1 + x^2 \leq (x\sin x + \cos x)$ for $x \in [0, \infty)$ (D) $1 + x^2 < (x \sin x + \cos x)$ for $x \in [0, \infty)$

MMN138

4. Which of the following is true

(A) $\sin x - \sin 2x \leq 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$ (B) $\sin x - \sin 2x \geq 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$

(C) $\sin x - \sin 2x \leq 2x$ for all $x \in \left[-\frac{\pi}{3}, 0\right]$ (D) $\sin x - \sin 2x > 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$

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MMN139

5. Which of the following is true

(A) $\frac{x^2}{2} + 2x + 3 > (3 - x)e^x$ for all $x \geq 0$ (B) $\frac{x^2}{2} + 2x + 3 < (3 - x)e^x$ for all $x \geq 0$

(C) $\frac{x^2}{2} + 2x + 3 \geq (3 - x)e^x$ for all $x \geq 0$ (D) $\frac{x^2}{2} + 2x + 3 \leq (3 - x)e^x$ for all $x \geq 0$

MMN140

Multiple Correct Question

6. If $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ where $0 < x < 1$, then

(A) $'f'$ is increasing function (B) $'g'$ is decreasing function

(C) $\frac{f(x)}{g(x)}$ is increasing function (D) $g(f(x))$ is decreasing function

MMN141

ANSWER KEY

Question	1	2	3	4	5	6
Answer	C	D	B	A	C	A,B,C

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Monotonicity : RACE-03

Single Correct Question

1. Exhaustive set of values of 'a' for which the function $f(x) = x^4 + \frac{3x^2}{2} + ax^3 + 1$ will be concave upward along the entire real line, is :

(A) [–1,1] (B) [–2,2] (C) [0,2] (D) [0,4]

MMN142

2. For which interval, the function $\frac{x^2-3x}{x-1}$ satisfies all the conditions of Rolle's theorem

(A) [0, 3] (B) [–3, 0] (C) [1.5, 3] (D) For no interval

MMN143

3. Rolle's theorem is not applicable to the function $f(x) = |x|$ defined on $[-1, 1]$ because

(A) f is not continuous on [–1, 1] (B) f is not differentiable on (–1,1)

(C) $f(-1) \neq f(1)$ (D) $f(-1) = f(1) \neq 0$

MMN144

4. The function $f(x) = x(x+3)e^{-(1/2)x}$ satisfies all the conditions of Rolle's theorem in [–3, 0]. The value of c is

(A) 0 (B) –1 (C) – 2 (D) – 3

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MMN145

5. Rolle's theorem is true for the function $f(x) = x^2 - 4$ in the interval

(A) [–2, 0] (B) [–2, 2] (C) $\left[0, \frac{1}{2}\right]$ (D) [0,2]

MMN146

6. For the function $f(x) = e^x,\ a = 0,\ b = 1$, the value of c in Lagrange's mean value theorem will be

(A) $\log x$ (B) $\log(e-1)$ (C) 0 (D) 1

MMN147

7. If $f(x) = \cos x, 0 \leq x \leq \frac{\pi}{2}$, then the real number '$c$' of the Lagrange's mean value theorem is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\sin^{-1}\left(\frac{2}{\pi}\right)$ (D) $\cos^{-1}\left(\frac{2}{\pi}\right)$

MMN148

8. From Lagrange's mean value theorem $f(b) - f(a) = (b-a)f'(x_1);\ a < x_1 < b$ if $f(x) = \frac{1}{x}$, then $x_1 =$

(A) $\sqrt{ab}$ (B) $\frac{a+b}{2}$ (C) $\frac{2ab}{a+b}$ (D) $\frac{b-a}{b+a}$

MMN149

9. For the function $x + \frac{1}{x}, x \in [1,3]$, the value of c for the Lagrange's mean value theorem is

(A) 1 (B) $\sqrt{3}$ (C) 2 (D) None of these

MMN150

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10. Let $f(x)$ satisfy all the conditions of Lagrange's mean value theorem in $[0, 2]$.
If $f(0) = 0$ and $|f'(x)| \leq \frac{1}{2}$ for all x, in $[0, 2]$ then
(A) $f(x) \leq 2$ (B) $|f(x)| \leq 1$
(C) $f(x) = 2x$ (D) $f(x) = 3$ for at least one x in $[0, 2]$

MMN151

11. In the Lagrange's mean value theorem $\frac{f(b)-f(a)}{b-a} = f'(c)$, if $a = 0, b = \frac{1}{2}$ and $f(x) = x(x-1)(x-2)$, the value of c is
(A) $1 - \frac{\sqrt{15}}{6}$ (B) $1 + \sqrt{15}$ (C) $1 - \frac{\sqrt{21}}{6}$ (D) $1 + \sqrt{21}$

MMN152

12. The abscissa of the points of the curve $y = x^3$ in the interval $[-2, 2]$, where the slope of the tangents can be obtained by Lagrange's mean value theorem for the interval $[-2, 2]$, are
(A) $\pm \frac{2}{\sqrt{3}}$ (B) $\pm\sqrt{3}$ (C) $\pm \frac{\sqrt{3}}{2}$ (D) 0

MMN153

13. The function $f(x) = (x-3)^2$ satisfies all the conditions of Lagrange's mean value theorem in $[3, 4]$. A point on $y = (x-3)^2$, where the tangent is parallel to the chord joining $(3, 0)$ and $(4, 1)$ is
(A) $\left(\frac{7}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (C) $(1, 4)$ (D) $(4, 1)$

MMN154

14. In the Lagrange's mean value theorem, $f(b) - f(a) = (b-a)f'(c)$ if $a = 4, b = 9$ and $f(x) = \sqrt{x}$ then the value of c is
(A) 8.00 (B) 5.25 (C) 4.00 (D) 6.25

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MMN155

15. Let Rolle's theorem holds for $f(x) = x^3 + bx^2 + ax$, when $1 \leq x \leq 2$ at the point $c = \frac{4}{3}$, then find $a + b$.
(A) 1 (B) 2 (C) 3 (D) 4

MMN156

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	B	C	B	B	C	A	B	B
Question	11	12	13	14	15					
Answer	C	A	B	D	C					

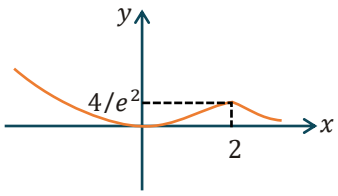
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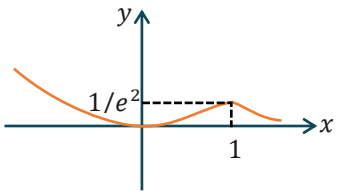
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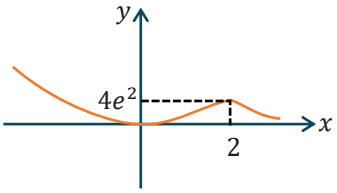
Monotonicity : RACE-04

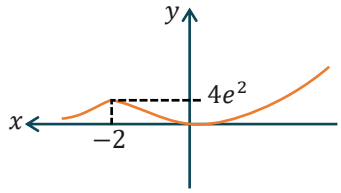
Single Correct Question

1. Draw graph of $y = x^2e^{-x}$.

(A) 

(B) 

(C) 

(D) 

MMN157

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Subjective Type Question

2. Draw graph of $f(x) = x\ell nx$

MMN158

3. Draw graph of $f(x)=\frac{x}{\ell nx}$

MMN159

4. Draw graph of $f(x) = \frac{2x-3}{x+1}$

MMN160

5. Draw graph of $f(x) = \frac{x^2-x+1}{x^2+x+1}$

MMN161

ANSWER KEY

Question	1
Answer	A

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